INCH-POUND

MIL-DTL-24308/2R
2 November 1999
SUPERSEDING
MIL-PRF-24308/2P
2 December 1996

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRIC, RECTANGULAR, MINIATURE, POLARIZED SHELL, RACK AND PANEL, RECEPTACLE, SOCKET CONTACTS, GENERAL PURPOSE, CLASS D AND G, CRIMP TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24308.

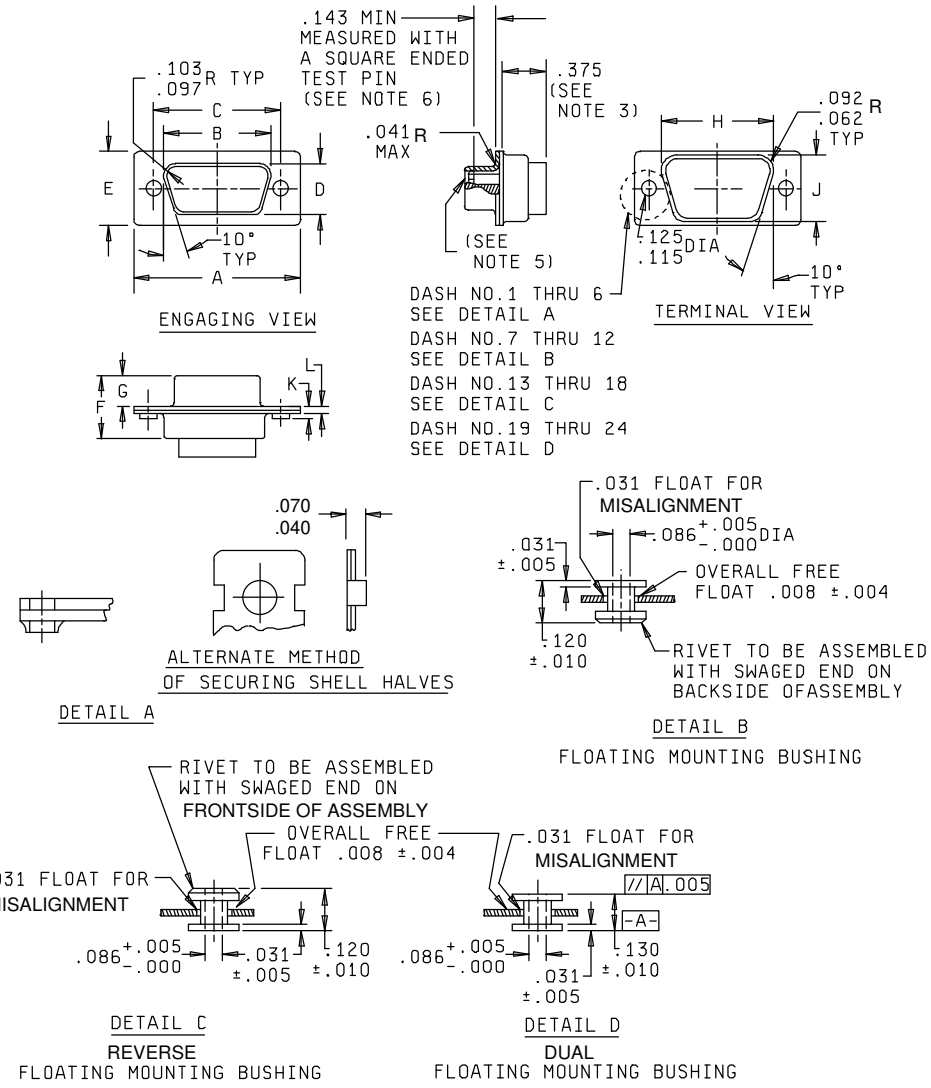


FIGURE 1. Shell, receptacle, crimp type

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Detail 1/	Shell size	Dimensions										
		A	B	C	D	E	F	G	H	J	K	L
A	1	1.228 (31.19) 1.198 (30.43)	.648 (16.46) .638 (16.21)	.989 (25.12) .979 (24.87)	.316 (8.03) .302 (7.77)	.509 (12.93) .479 (12.17)	.439 (11.15) .419 (10.54)	.248 (6.30) .238 (6.05)	.769 (19.53) .749 (19.02)	.432 (10.97) .412 (10.46)	.060 (1.52) .035 (.89)	.040 (1.02) .020 (.51)
	2	1.556 (39.52) 1.526 (38.76)	.976 (24.79) .966 (24.79)	1.317 (33.45) 1.307 (33.20)	.316 (8.03) .302 (7.77)	.509 (12.93) .479 (12.17)	.439 (11.15) .419 (10.54)	.248 (6.30) .238 (6.05)	1.093 (27.76) 1.073 (27.25)	.432 (10.97) .412 (10.46)	.060 (1.52) .035 (.89)	.040 (1.02) .020 (.51)
	3	2.103 (53.42) 2.073 (52.65)	1.516 (38.51) 1.506 (38.25)	1.857 (47.17) 1.847 (46.91)	.316 (8.03) .302 (7.77)	.509 (12.93) .479 (12.17)	.439 (11.15) .419 (10.54)	.248 (6.30) .238 (6.05)	1.635 (41.53) 1.615 (41.02)	.432 (10.97) .412 (10.46)	.060 (1.52) .035 (.89)	.049 (1.25) .029 (.74)
	4	2.744 (69.76) 2.714 (68.94)	2.164 (54.97) 2.154 (54.71)	2.505 (63.63) 2.495 (63.37)	.316 (8.03) .302 (7.77)	.509 (12.93) .479 (12.17)	.439 (11.15) .419 (10.54)	.248 (6.30) .238 (6.05)	2.282 (57.96) 2.262 (57.45)	.432 (10.97) .412 (10.46)	.060 (1.52) .035 (.89)	.049 (1.25) .029 (.74)
	5	2.650 (67.31) 2.620 (66.55)	2.069 (52.55) 2.059 (52.30)	2.411 (61.24) 2.401 (60.99)	.428 (10.87) .418 (10.62)	.620 (15.75) .590 (14.99)	.439 (11.15) .419 (10.54)	.248 (6.30) .238 (6.05)	2.188 (55.58) 2.168 (55.07)	.544 (13.82) .524 (13.31)	.060 (1.52) .035 (.89)	.049 (1.25) .029 (.74)
	6	2.744 (69.70) 2.714 (68.94)	2.194 (55.73) 2.184 (55.47)	2.505 (63.63) 2.495 (63.37)	.490 (12.45) .480 (12.19)	.683 (17.35) .653 (16.59)	.439 (11.15) .419 (10.54)	.248 (6.30) .238 (6.05)	2.312 (58.72) 2.292 (58.22)	.606 (15.39) .586 (14.88)	.060 (1.52) .035 (.89)	.049 (1.25) .029 (.74)
B 2/	1	Same as detail A, shell size 1 except in accordance with detail B.										
	2	Same as detail A, shell size 2 except in accordance with detail B.										
	3	Same as detail A, shell size 3 except in accordance with detail B.										
	4	Same as detail A, shell size 4 except in accordance with detail B.										
	5	Same as detail A, shell size 5 except in accordance with detail B.										
	6	Same as detail A, shell size 6 except in accordance with detail B.										
C 2/	1	Same as detail A, shell size 1 except in accordance with detail C.										
	2	Same as detail A, shell size 2 except in accordance with detail C.										
	3	Same as detail A, shell size 3 except in accordance with detail C.										
	4	Same as detail A, shell size 4 except in accordance with detail C.										
	5	Same as detail A, shell size 5 except in accordance with detail C.										
	6	Same as detail A, shell size 6 except in accordance with detail C.										
D	1	Same as detail A, shell size 1 except in accordance with detail D.										
	2	Same as detail A, shell size 2 except in accordance with detail D.										
	3	Same as detail A, shell size 3 except in accordance with detail D.										
	4	Same as detail A, shell size 4 except in accordance with detail D.										
	5	Same as detail A, shell size 5 except in accordance with detail D.										
	6	Same as detail A, shell size 6 except in accordance with detail D.										

1/ See table II for supersession data.
2/ Details B and C are inactive for new design, use detail D.

FIGURE 1. Shell, receptacle, crimp type - Continued.

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NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Insulator may project .375 (9.53) maximum.
4. Wire insulation .072 (1.83) maximum for arrangements, (see MIL-DTL-24308, Appendix A), A-1-1, A-2-1, A-3-1, A-4-1,and A-5-1. Wire insulation .054 (1.37) maximum for arrangements, (see MIL-DTL-24308, Appendix A),A-1-2, A-2-2, A-3-2, A-4-2, A-5-2, and A-6-1.
5. To accommodate a .040 (1.02) diameter pin for arrangements, (see MIL-DTL-24308, Appendix A), A-1-1, A-2-1, A-3-1, A-4-1, and A-5-1. To accommodate a .030 (.76) diameter pin for arrangements, (see MIL-PRF-24308, Appendix A),A-1-2, A-2-2, A-3-2, A-4-2, A-5-2, and A-6-1
6. Use .041/.039 (1.04/.99) diameter pin for arrangements, (see MIL-DTL-24308, Appendix A),A-1.1, A-2.1, A-3.1, A-4.1,and A-5.1. Use .0305/.0295 (.77/.75) diameter pin for arrangements, (see MIL-DTL-24308, Appendix A), A-1-2, A-2-2, A-3-2, A-4-2, A-5-2, and A-6-1
7. Patent notice. The government has a royalty free license under the U.S. Patents listed below for the benefit of manufacturers of the items disclosed and described herein, either for the government of for use in equipment delivered to the government.

2790153 expires April 23, 1974
3110093 expires November 12, 1980
3158424 expires November 24, 1981

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Inches	mm
.004	.10
.005	.13
.008	.20
.010	.25
.031	.79
.040	1.02
.041	1.04
.062	1.57
.070	1.78
.086	2.18
.092	2.34
.097	2.46
.103	2.62
.115	2.92
.120	3.05
.125	3.18
.129	3.28
.143	3.63
.375	9.53

FIGURE 1. Shell, receptacle, crimp type - Continued.

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REQUIREMENTS:

The connectors shall conform to figure 1 and the specification Part or Identifying Numbers (PIN) indicated in table I.

TABLE I. Design standards. 1/

(PIN) Class G	PIN Class D	Shell size	Shell detail (see figure1)	Insert arrangement reference number 6/	Contact PIN	PIN marked on connector class G	PIN marked on connector class D
M24308/2-1	M24308/2-516	1	A	A-1-1	M39029/63-368	M24308/2-1	M24308/2-516
M24308/2-2	M24308/2-517	2	A	A-2-1	M39029/63-368	M24308/2-2	M24308/2-517
M24308/2-3	M24308/2-518	3	A	A-3-1	M39029/63-368	M24308/2-3	M24308/2-518
M24308/2-4	M24308/2-519	4	A	A-4-1	M39029/63-368	M24308/2-4	M24308/2-519
M24308/2-5	M24308/2-520	5	A	A-5-1	M39029/63-368	M24308/2-5	M24308/2-520
M24308/2-6	M24308/2-521	Connector kit: M24308/2-1 connector with applicable tool 2/				M24308/2-1	M24308/2-516
M24308/2-7	M24308/2-522	Connector kit: M24308/2-2 connector with applicable tool 2/				M24308/2-2	M24308/2-517
M24308/2-8	M24308/2-523	Connector kit: M24308/2-3 connector with applicable tool 2/				M24308/2-3	M24308/2-518
M24308/2-9	M24308/2-524	Connector kit: M24308/2-4 connector with applicable tool 2/				M24308/2-4	M24308/2-519
M24308/2-10	M24308/2-525	Connector kit: M24308/2-5 connector with applicable tool 2/				M24308/2-5	M24308/2-520
M24308/2-11	M24308/2-526	1	A	A-1-2	M39029/57-354	M24308/2-11	M24308/2-526
M24308/2-12	M24308/2-527	2	A	A-2-2	M39029/57-354	M24308/2-12	M24308/2-527
M24308/2-13	M24308/2-528	3	A	A-3-2	M39029/57-354	M24308/2-13	M24308/2-528
M24308/2-14	M24308/2-529	4	A	A-4-2	M39029/57-354	M24308/2-14	M24308/2-529
M24308/2-15	M24308/2-530	5	A	A-5-2	M39029/57-354	M24308/2-15	M24308/2-530
M24308/2-16	M24308/2-531	6	A	A-6-1	M39029/57-354	M24308/2-16	M24308/2-531
M24308/2-17	M24308/2-532	Connector kit: M24308/2-11 connector with applicable tool 3/				M24308/2-11	M24308/2-526
M24308/2-18	M24308/2-533	Connector kit: M24308/2-12 connector with applicable tool 3/				M24308/2-12	M24308/2-527
M24308/2-19	M24308/2-534	Connector kit: M24308/2-13 connector with applicable tool 3/				M24308/2-13	M24308/2-528
M24308/2-20	M24308/2-535	Connector kit: M24308/2-14 connector with applicable tool 3/				M24308/2-14	M24308/2-529
M24308/2-21	M24308/2-536	Connector kit: M24308/2-15 connector with applicable tool 3/				M24308/2-15	M24308/2-530
M24308/2-22	M24308/2-537	Connector kit: M24308/2-16 connector with applicable tool 3/				M24308/2-16	M24308/2-531

See footnotes at end of table.

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TABLE I. Design standards - Continued. 1/

PIN class G	PIN class D	Shell size	Shell detail (see figure 1)	Insert arrangement reference number 6/ .	Contact PIN	PIN marked on connector class G	PIN marked on connector class D
4/ M24308/2-23	4/ M24308/2-538	1	B	A-1-1	M39029/63-368	M24308/2-23	M24308/2-538
M24308/2-24	M24308/2-539	2	B	A-2-1	M39029/63-368	M24308/2-24	M24308/2-539
M24308/2-25	M24308/2-540	3	B	A-3-1	M39029/63-368	M24308/2-25	M24308/2-540
M24308/2-26	M24308/2-541	4	B	A-4-1	M39029/63-368	M24308/2-26	M24308/2-541
M24308/2-27	M24308/2-542	5	B	A-5-1	M39029/63-368	M24308/2-27	M24308/2-542
M24308/2-28	M24308/2-543	1	B	A-1-2	M39029/57-354	M24308/2-28	M24308/2-543
M24308/2-29	M24308/2-544	2	B	A-2-2	M39029/57-354	M24308/2-29	M24308/2-544
M24308/2-30	M24308/2-545	3	B	A-3-2	M39029/57-354	M24308/2-30	M24308/2-545
M24308/2-31	M24308/2-546	4	B	A-4-2	M39029/57-354	M24308/2-31	M24308/2-546
M24308/2-32	M24308/2-547	5	B	A-5-2	M39029/57-354	M24308/2-32	M24308/2-547
M24308/2-33	M24308/2-548	6	B	A-6-1	M39029/57-354	M24308/2-33	M24308/2-548
4/ M24308/2-34	M24308/2-549	Connector kit: M24308/2-23 connector with applicable tool 2/				M24308/2-23	M24308/2-549
M24308/2-35	M24308/2-550	Connector kit: M24308/2-24 connector with applicable tool 2/				M24308/2-24	M24308/2-550
M24308/2-36	M24308/2-551	Connector kit: M24308/2-25 connector with applicable tool 2/				M24308/2-25	M24308/2-551
M24308/2-37	M24308/2-552	Connector kit: M24308/2-26 connector with applicable tool 2/				M24308/2-26	M24308/2-552
M24308/2-38	M24308/2-553	Connector kit: M24308/2-27 connector with applicable tool 2/				M24308/2-27	M24308/2-553
M24308/2-39	M24308/2-554	Connector kit: M24308/2-28 connector with applicable tool 2/				M24308/2-28	M24308/2-554
M24308/2-40	M24308/2-555	Connector kit: M24308/2-29 connector with applicable tool 2/				M24308/2-29	M24308/2-555
M24308/2-41	M24308/2-556	Connector kit: M24308/2-30 connector with applicable tool 2/				M24308/2-30	M24308/2-556
M24308/2-42	M24308/2-557	Connector kit: M24308/2-31 connector with applicable tool 2/				M24308/2-31	M24308/2-557
M24308/2-43	M24308/2-558	Connector kit: M24308/2-32 connector with applicable tool 2/				M24308/2-32	M24308/2-558
M24308/2-44	M24308/2-559	Connector kit: M24308/2-33 connector with applicable tool 2/				M24308/2-33	M24308/2-559

See footnotes at end of table.

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TABLE I. Design standards - Continued. 1/

PIN class G	PIN class D	Shell size	Shell detail (see figure 1)	Insert arrangement reference number 6/	Contact PIN	PIN marked on connector class G	PIN marked on connector class D
M24308/2-281	M24308/2-560	1	A	A-1-1	Contacts not included	M24308/2-1	M24308/2-516
M24308/2-282	M24308/2-561	2	A	A-2-1	"	M24308/2-2	M24308/2-517
M24308/2-283	M24308/2-562	3	A	A-3-1	"	M24308/2-3	M24308/2-518
M24308/2-284	M24308/2-563	4	A	A-4-1	"	M24308/2-4	M24308/2-519
M24308/2-285	M24308/2-564	5	A	A-5-1	"	M24308/2-5	M24308/2-520
M24308/2-286	M24308/2-565	1	A	A-1-2	"	M24308/2-11	M24308/2-526
M24308/2-287	M24308/2-566	2	A	A-2-2	"	M24308/2-12	M24308/2-527
M24308/2-288	M24308/2-567	3	A	A-3-2	"	M24308/2-13	M24308/2-528
M24308/2-289	M24308/2-568	4	A	A-4-2	"	M24308/2-14	M24308/2-529
M24308/2-290	M24308/2-569	5	A	A-5-2	"	M24308/2-15	M24308/2-530
M24308/2-291	M24308/2-570	6	A	A-6-1	"	M24308/2-16	M24308/2-531
4/ M24308/2-292	4/ M24308/2-571	1	B	A-1-1	"	M24308/2-23	M24308/2-538
M24308/2-293	M24308/2-572	2	B	A-2-1	"	M24308/2-24	M24308/2-539
M24308/2-294	M24308/2-573	3	B	A-3-1	"	M24308/2-25	M24308/2-540
M24308/2-295	M24308/2-574	4	B	A-4-1	"	M24308/2-26	M24308/2-541
M24308/2-296	M24308/2-575	5	B	A-5-1	"	M24308/2-27	M24308/2-542
M24308/2-297	M24308/2-576	1	B	A-1-2	"	M24308/2-28	M24308/2-543
M24308/2-298	M24308/2-577	2	B	A-2-2	"	M24308/2-29	M24308/2-544
M24308/2-299	M24308/2-578	3	B	A-3-2	"	M24308/2-30	M24308/2-545
M24308/2-300	M24308/2-579	4	B	A-4-2	"	M24308/2-31	M24308/2-546
M24308/2-301	M24308/2-580	5	B	A-5-2	"	M24308/2-32	M24308/2-547
M24308/2-302	M24308/2-581	6	B	A-6-1	"	M24308/2-33	M24308/2-548
4/ M24308/2-335	M24308/2-582	3	C	A-3-1	M39029/63-368	M24308/2-335	M24308/2-582
M24308/2-336	M24308/2-583	4	C	A-4-1	M39029/63-368	M24308/2-336	M24308/2-583
M24308/2-341	M24308/2-584	2	C	A-2-1	M39029/63-368	M24308/2-341	M24308/2-584
M24308/2-342	M24308/2-585	1	D	A-1-1	M39029/63-368	M24308/2-342	M24308/2-585
M24308/2-343	M24308/2-586	2	D	A-2-1	M39029/63-368	M24308/2-343	M24308/2-586
M24308/2-344	M24308/2-587	3	D	A-3-1	M39029/63-368	M24308/2-344	M24308/2-587
M24308/2-345	M24308/2-588	4	D	A-4-1	M39029/63-368	M24308/2-345	M24308/2-588
M24308/2-346	M24308/2-589	5	D	A-5-1	M39029/63-368	M24308/2-346	M24308/2-589
M24308/2-347	M24308/2-590	1	D	A-1-2	M39029/57-354	M24308/2-347	M24308/2-590
M24308/2-348	M24308/2-591	2	D	A-2-2	M39029/57-354	M24308/2-348	M24308/2-591
M24308/2-349	M24308/2-592	3	D	A-3-2	M39029/57-354	M24308/2-349	M24308/2-592
M24308/2-350	M24308/2-593	4	D	A-4-2	M39029/57-354	M24308/2-350	M24308/2-593
M24308/2-351	M24308/2-594	5	D	A-5-2	M39029/57-354	M24308/2-351	M24308/2-594
M24308/2-352	M24308/2-595	6	D	A-6-1	M39029/57-354	M24308/2-352	M24308/2-595

See footnotes at end of table.

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TABLE I. Design standards - Continued. 1/

PIN class G	PIN class D	Shell size	Shell detail (see figure 1)	Insert arrangement reference number 6/	Contact PIN	PIN marked on connector class G	PIN marked on connector class D
M24308/2-353	M24308/2-596	Connector kit: M24308/2-342 connector with applicable tool 2/				M24308/2-342	M24308/2-585
M24308/2-354	M24308/2-597	Connector kit: M24308/2-343 connector with applicable tool 2/				M24308/2-343	M24308/2-586
M24308/2-355	M24308/2-598	Connector kit: M24308/2-344 connector with applicable tool 2/				M24308/2-344	M24308/2-587
M24308/2-356	M24308/2-599	Connector kit: M24308/2-345 connector with applicable tool 2/				M24308/2-345	M24308/2-588
M24308/2-357	M24308/2-600	Connector kit: M24308/2-346 connector with applicable tool 2/				M24308/2-346	M24308/2-589
M24308/2-358	M24308/2-601	Connector kit: M24308/2-347 connector with applicable tool 2/				M24308/2-347	M24308/2-590
M24308/2-359	M24308/2-602	Connector kit: M24308/2-348 connector with applicable tool 2/				M24308/2-348	M24308/2-591
M24308/2-360	M24308/2-603	Connector kit: M24308/2-349 connector with applicable tool 2/				M24308/2-349	M24308/2-592
M24308/2-361	M24308/2-604	Connector kit: M24308/2-350 connector with applicable tool 2/				M24308/2-350	M24308/2-593
M24308/2-362	M24308/2-605	Connector kit: M24308/2-351 connector with applicable tool 2/				M24308/2-351	M24308/2-594
M24308/2-363	M24308/2-606	Connector kit: M24308/2-352 connector with applicable tool 2/				M24308/2-352	M24308/2-595
M24308/2-482	M24308/2-607	1	D	A-1-1	Contacts not included	M24308/2-342	M24308/2-585
M24308/2-483	M24308/2-608	2	D	A-2-1	"	M24308/2-343	M24308/2-586
M24308/2-484	M24308/2-609	3	D	A-3-1	"	M24308/2-344	M24308/2-587
M24308/2-485	M24308/2-610	4	D	A-4-1	"	M24308/2-345	M24308/2-588
M24308/2-486	M24308/2-611	5	D	A-5-1	"	M24308/2-346	M24308/2-589
M24308/2-487	M24308/2-612	1	D	A-1-2	"	M24308/2-347	M24308/2-590
M24308/2-488	M24308/2-613	2	D	A-2-2	"	M24308/2-348	M24308/2-591
M24308/2-489	M24308/2-614	3	D	A-3-2	"	M24308/2-349	M24308/2-592
M24308/2-490	M24308/2-615	4	D	A-4-2	"	M24308/2-350	M24308/2-593
M24308/2-491	M24308/2-616	5	D	A-5-2	"	M24308/2-351	M24308/2-594
M24308/2-492	M24308/2-617	6	D	A-6-1	"	M24308/2-352	M24308/2-595
5/ M24308/2-509	5/ M24308/2-621	3	D	A-3-1	M39029/63-368		M24308/2-621
M24308/2-510	M24308/2-622	4	D	A-4-1	M39029/63-368		M24308/2-622
M24308/2-515	M24308/2-627	2	D	A-2-1	M39029/63-368		M24308/2-627

1/ These connectors mate with connectors specified in MIL-DTL- 24308/3, MIL-DTL-24308/4, MIL-DTL-24308/9, and MIL-DTL-24308/24.

2/ Installation and removal tool: M81969/39-01 with plastic tips or M81969/1-02 with metal tips. For size 20 contacts, either tool shall be supplied with each individual connector.

3/ Installation and removal tool: M81969/14-01 with plastic tips or M81969/1-04 with metal tips. For size 22 contacts, either tool shall be supplied with each individual connector.

4/ PINs M24308/2-23 through M24308/2-44, M24308/2-538 through M24308/2-559 and M24308/2-292 through M24308/2-292 through M24308/2-341, M24308/2-571 through M24308/2-584 are inactive for new design. Use M24308/2-342 through M24308/2-352, and M24308/2-585 through M24308/2-595.

5/ PINs M24308/2-509, M24308/2-510, M24308/2-515, M24308/2-621, M24308/2-622, and M24308/2-627are inactive for new design. Use M24308/2-344, M24308/2-345, M24308/2-343, M24308/2-587, M24308/2-588, and M24308/2-586, respectively.

6/. For insert arrangements see MIL-DTL-24308 Appendix A.

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Supersession data: See table II.

TABLE II. Shell supersession data.

Shell detail (see figure 1)	Superseded MS sheet
Detail A	MS18270-1 through -6
Detail B	MS18270-7 through -12
Detail C	MS18270-13 through -18
Detail D	MS18270-19 through -24

PIN: Consists of the letter M, the basic number of the specification sheet, and a dash number compiled from the code.

PIN example: M24308/2 -361 F

Specification sheet number

Data sheet number from table I

Finish: For class G:
F = Cadmium.
No suffix = Zinc.

For class D:
Class D connectors have a nickel finish, no suffix is required.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Army - AT, AV, CR-4, MI
Navy - AS, CG, MC, SH
Air Force - 99

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